# Jurong Shipyard Pte Ltd

*(Regn. No. 199908265G)*
29 TANJONG KLING ROAD
SINGAPORE 628054

JSL

**Serial No.** 071289/A

## WELDER AND WELDING OPERATOR QUALIFICATION TEST RECORD

Welder [X] Welding Operator [--] Tack Welder [--]

Name: MOHAMMED RAJAN MOHAMMED HALIM Identification: 062445343 Badge No.: 14753

WPS No.: SMAW-S010-02 Welding Process: SMAW Subcon/Section: YKE/JTS

Supporting PQR: SS-SMSM-GRP02-010 Type of Process: Manual [X] Machine [--] Semi-Auto [--] Auto [--]

| | | Actual | AWS Range | | ASME Range | |
|---|---|---|---|---|---|---|
| Position | F / H / V Up / V Down / OH / All / TYK | 3G Vertical Up | F,H,V Up | F,H,V Up | F,V Up | F,H,V Up |
| Type of Joint | Groove (G) / Fillet (F) | GROOVE | Groove | Fillet | Groove | Fillet |
| Base Metal | Thickness (mm) | 25.40 | 1.5mm to unlimited | 1.5mm to unlimited | Unlimited | Unlimited |
| | Diameter (inches) | N.A. | Above 24.0in | Unlimited | Above 24.0in | Unlimited |
| | Material Specification | ABS | Not Applicable | | ABS EH36 | |
| | Type / Grade | EH36 | | | | |
| | P No / Group No | N.A./GRPII | | | | |
| Filler Metal | AWS Specification | A5.1 | Not Applicable | | | |
| | AWS Classification | E7016 | | | | |
| | F No | F4 | F1,F2,F3,F4 | N.A. | F1,F2,F3,F4 | N.A. |
| Joint Backing | With / Without | With | With* | Without* | With* | Without* |
| Shielding Gas | With / Without Backing | N.A. | N.A. | | N.A. | |
| Electrical Characteristic | Current or Polarity | A.C. | Not Applicable | | N.A. | |
| | Transfer Mode | N.A. | | | N.A. | |

### Visual Inspection

| Appearance | SATISFACTORY | Undercut | NIL | Piping Porosity | NIL |
|---|---|---|---|---|---|

### Guided Bend Test Result

| Type | Result | Type | Result |
|---|---|---|---|
| N.A. | N.A. | N.A. | N.A. |
| N.A. | N.A. | N.A. | N.A. |

### Fillet Test Results

| Appearance | N.A. | Fillet Size | N.A. | Macro-etch | N.A. |
|---|---|---|---|---|---|

### Fracture Test

| Root Penetration | N.A. |
|---|---|

(Describe the location, nature and size of any crack or tearing of specimen)

Test Conducted By: N.A. Organization: N.A. Test Date: N.A.

Laboratory Test No.: N.A. Test Witness by: (Yard/Owner/Class) N.A.

### Radiography Test Results

| Film Identification | Result | Remarks | Film Identification | Result | Remarks |
|---|---|---|---|---|---|
| 062445343 | SATISFACTORY | 3G | N.A. | N.A. | N.A. |
| N.A. | N.A. | N.A. | N.A. | N.A. | N.A. |

Test Conducted By: CHONG Organization: ROYSCAN NDT Test Date: 31 Aug 2007

RT Report No.: RT/070035-991 Test Witness by: (Yard/Owner/Class) S RETHINA SABAPATHI/SONG XIAO JUN

We, the undersigned, certified that the statements in this record are correct and that the welds are prepared and tested in accordance with the requirements of

[X] Structure Welding Code-Steel. Section 4C of AWS D1.1, Year 2006

[X] Boiler and Pressure Vessel Code. Section IX of ASME Code, Year 2004

Authorized by:

JURONG SHIPYARD PTE LTD QA / QC

Chia Meng Koon
QA/QC Manager

Endorsed by:

AMERICAN BUREAU OF SHIPPING SINGAPORE

SONG XIAO JUN
ABS Surveyor

# Jurong Shipyard Pte Ltd

*(Regn. No. 199908265G)*
29 TANJONG KLING ROAD
SINGAPORE 628054

JSL

**Serial No.** 083068/V

## WELDER AND WELDING OPERATOR QUALIFICATION TEST RECORD

Welder [X] Welding Operator [--] Tack Welder [--]

Name: MOHAMMED RAJAN MOHAMMED HALIM Identification: 062445343 Badge No.: 14753

WPS No.: SMAW-S007-02 Welding Process: SMAW Subcon/Section: YKE/JIS

Supporting PQR: SS-SMSM-GRP02-007 Type of Process: Manual [X] Machine [--] Semi-Auto [--] Auto [--]

| | | Actual | AWS Range | | ASME Range | |
|---|---|---|---|---|---|---|
| Position | F / H / V Up / V Down / OH / All / TYK | 4G Overhead | F,OH | F,H,OH | F,OH | F,H,OH |
| Type of Joint | Groove (G) / Fillet (F) | GROOVE | Groove | Fillet | Groove | Fillet |
| Base Metal | Thickness (mm) | 25.40 | 3.0mm to unlimited | 3.0mm to unlimited | Unlimited | Unlimited |
| | Diameter (inches) | N.A. | Above 24.0in | Unlimited | Above 24.0in | Unlimited |
| | Material Specification | ABS | Not Applicable | | ABS EH36 | |
| | Type / Grade | EH36 | | | | |
| | P No / Group No | N.A./GRPII | | | | |
| Filler Metal | AWS Specification | A5.1 | Not Applicable | | | |
| | AWS Classification | E7016 | | | | |
| | F No | F4 | F1,F2,F3,F4 | N.A. | F1,F2,F3,F4 | N.A. |
| Joint Backing | With / Without | With | With* | Without* | With* | Without* |
| Shielding Gas | With / Without Backing | N.A. | N.A. | | N.A. | |
| Electrical Characteristic | Current or Polarity | DCEP | Not Applicable | | N.A. | |
| | Transfer Mode | N.A. | | | N.A. | |

### Visual Inspection

| | | | | | |
|---|---|---|---|---|---|
| Appearance | SATISFACTORY | Undercut | NIL | Piping Porosity | NIL |

### Guided Bend Test Result

| Type | Result | Type | Result |
|---|---|---|---|
| N.A. | N.A. | N.A. | N.A. |
| N.A. | N.A. | N.A. | N.A. |

### Fillet Test Results

| | | | | | |
|---|---|---|---|---|---|
| Appearance | N.A. | Fillet Size | N.A. | Macro-etch | N.A. |

### Fracture Test

| | |
|---|---|
| Root Penetration | N.A. |

(Describe the location, nature and size of any crack or tearing of specimen)

Test Conducted By: N.A. Organization: N.A. Test Date: N.A.

Laboratory Test No.: N.A. Test Witness by: (Yard/Owner/Class) N.A.

### Radiography Test Results

| Film Identification | Result | Remarks | Film Identification | Result | Remarks |
|---|---|---|---|---|---|
| 062445343 | SATISFACTORY | 4G | N.A. | N.A. | N.A. |
| N.A. | N.A. | N.A. | N.A. | N.A. | N.A. |

Test Conducted By: CHONG Organization: ROYSCAN NDT Test Date: 13 Aug 2008

RT Report No.: RT/080035-1527 Test Witness by: (Yard/Owner/Class) S RETHINA SABAPATHI/JOSEPH LIM

We, the undersigned, certified that the statements in this record are correct and that the welds are prepared and tested in accordance with the requirements of

[X] Structure Welding Code-Steel.
Section 4C of AWS D1.1, Year 2006

[X] Boiler and Pressure Vessel Code.
Section IX of ASME Code, Year 2007

Authorized by:

JURONG SHIPYARD PTE QA / QC

Chia Meng Koon
QA/QC Manager

Endorsed by:

AMERICAN BUREAU OF SHIPPING

22/09/08

JOSEPH LIM
ABS Surveyor

# Jurong Shipyard Pte Ltd

(Regn. No. 199908265G)
29 TANJONG KLING ROAD
SINGAPORE 628054

JSL

Serial No 093076/y

## WELDER AND WELDING OPERATOR QUALIFICATION TEST RECORD

Name: RAJAN MOHAMMED Identification: 0 62445343 Badge No: 17322

WPS No.: SMAW-S0145-01 Welding Process: SMAW Subcon/Section: HRE/HSS

Supporting PQR: SS-SMSM-GRP02-145 Type of Process: Manual [X] Machine [--] Semi-Auto [--] Auto [--]

| | | Actual | AWS Range | | ASME Range | |
|---|---|---|---|---|---|---|
| Position | F / H / V Up / V Down / OH / All / TYK | 6GR TYK | All,TYK | All,TYK | N.A. | N.A. |
| Type of Joint | Groove (G) / Fillet (F) | GROOVE | Groove | Fillet | Groove | Fillet |
| Base Metal | Thickness (mm) | 18.30 | 5.0mm to unlimited | 3.0mm to unlimited | N.A | N.A. |
| | Diameter (inches) | 6.00 | 4.0in to unlimited | Unlimited | N.A | N.A. |
| | Material Specification | API 5L | Not Applicable | | N.A. | |
| | Type / Grade | X52 | | | | |
| | Group No / P No | GRPII/P1G1 | | | | |
| Filler Metal | AWS Specification | A5.1 | Not Applicable | | | |
| | AWS Classification | E7016 | | | | |
| | F No | F4 | F1,F2,F3,F4 | F1,F2,F3,F4 | N.A | N.A |
| Joint Backing | With / Without | Without | With* | Without* | With* | Without* |
| Shielding Gas | With / Without Backing | N.A. | N.A. | | N.A. | |
| Electrical Characteristic | Current or Polarity | DCEP | Not Applicable | | N.A. | |
| | Transfer Mode | N.A. | | | N.A. | |

### Visual Inspection

| Appearance | SATISFACTORY | Undercut | NIL | Piping Porosity | NIL |
|---|---|---|---|---|---|

### Guided Bend Test Result

| Type | Result | Type | Result |
|---|---|---|---|
| N.A. | N.A. | N.A. | N.A. |
| N.A. | N.A. | N.A. | N.A. |

### Fillet Test Results

| Appearance | N.A. | Fillet Size | N.A. | Macro-etch | N.A. |
|---|---|---|---|---|---|

### Fracture Test

| Root Penetration | N.A. |
|---|---|

(Describe the location, nature and size of any crack or tearing of specimen)

Test Conducted By: N.A. Organization: N.A. Test Date: N.A.

Laboratory Test No.: N.A. Test Witness by: (Yard/Owner/Class) N.A.

### Radiography Test Results

| Film Identification | Result | Remarks | Film Identification | Result | Remarks |
|---|---|---|---|---|---|
| 0 62445343 | SATISFACTORY | 6GR | N.A. | N.A. | N.A. |
| N.A. | N.A. | N.A. | N.A. | N.A. | N.A. |

Test Conducted By: CHONG Organization: ROYSCAN NDT Test Date: 07 Sep 2011

RT Report No.: RT/110066-2433 Test Witness by: (Yard/Owner/Class) S RETHINA SABAPATHI/G R REDDY

We, the undersigned, certified that the statements in this record are correct and that the welds are prepared and tested in accordance with the requirements of

[X] Structure Welding Code-Steel Section 4C of AWS D1.1, Year 2010

[-] Boiler and Pressure Vessel Code Section IX of ASME Code, Year

Authorized by: JURONG SHIPYARD PTE LTD QA / QC

Chia Meng Koon
QA/QC Manager

Endorsed by: ABS SINGAPORE

14/09/2011

G R REDDY
ABS Surveyor



Plate - A